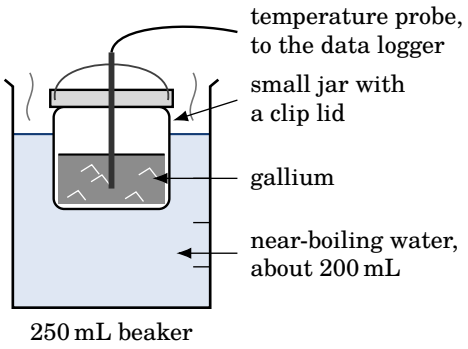
Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

Gallium Melting — Demo

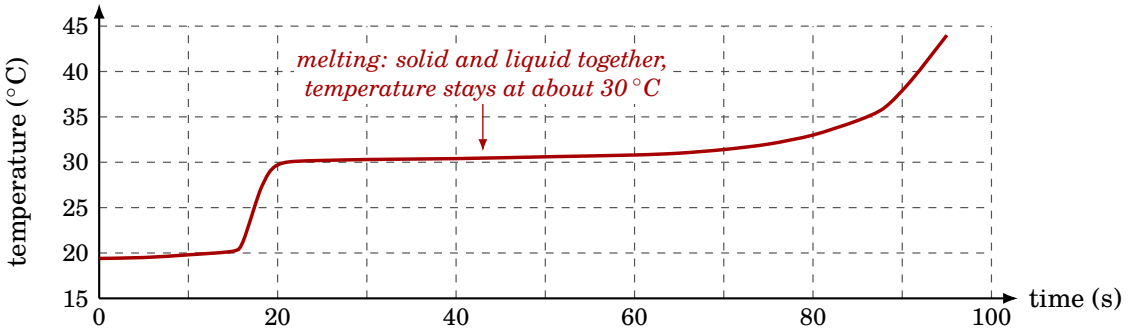


What you are watching

- **Gallium** is a metal that **melts at 29.8 °C**: above room temperature, below body temperature.
- The **probe** sits inside the gallium, so the graph on the screen is **the gallium's temperature**.
- Hot water heats the jar from outside. Watch the **metal** and the **graph** together.

Safety. The water is near boiling. Gallium stays sealed in its jar: it stains skin and damages aluminium.

Sketch the graph from the screen as it builds. Mark where the gallium is melting.



	Before the flat part	On the flat part	After the flat part
What it looks like	*a dull silver solid, in crystals*	*solid crystals with shiny liquid between them*	*a shiny silver liquid that moves when the jar moves*
State	*solid*	*solid and liquid: melting*	*liquid*

1) **Name** the change of state you saw, and give the temperature at which it happened. **[2]**

Melting (solid to liquid), at about 30 °C (gallium's melting point is 29.8 °C).

2) **Describe** what the temperature did while the gallium melted, and once it had all melted. **[2]**

While it melted, the temperature stayed almost the same, at about 30 °C (the flat part). Once it was all liquid, the temperature rose again.

3) **Explain** why melting is a physical change, and state what happens to the mass. **[2]**

No new substance is made: left to cool, the liquid turns back into solid gallium with its original properties. The mass stays the same, because the number of particles does not change.